

Manual

ERP Interface with Additional Data EIS-ERP / EIS-EZI 8.1

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1 MF Schnittstelle MES <=> ERP

Use options

The EIS-ERP interface is used to connect MES systems to superior ERP systems.

This covers the following general requirements:

- Default interface for the adoption of order sequencing data from the ERP system and confirmation/upload of recorded postings in the production.
- Rollout-capable solution.
- Close interfacing of HYDRA and PPS in order to account for the increasing sensitivity of the integrated solution.
- Maintenance options and administration of the interface.

This document describes the principles of the data exchange on the logical level and the system components that are required to this end. In addition, it offers a detailed overview of the interface's structure.

Implementation notes

Use the interface if

- you wish to automate the exchange of order and operation data between ERP and MES;
- you wish to automate the exchange of confirmations recorded in the MES to the ERP.

Integration

The interface constitutes the basis for the orders and operations administered in BDE.

And the data recorded in the BDE/shop floor data collection module in turn, constitute the basis for the confirmation/upload to the ERP system.

Scope of functions

- Transfer of order data
 - Interface enabling the communication between HYDRA and ERP/ PPS systems or external control stations/shop floor scheduling systems.

- Used to interface HYDRA with SAP R/3 PP, Baan IV, Baan ERP, ProAlpha, M-Suite (Soft-M), Brain XPPS, Brain AS, Oracle, PSI PENTA, Navision, ABAS ERP, SYS APS, Infor, IFR ERP, MS-Business Solutions, Axapta, MAPICS, MAS90, Orgaplan, FOSS (ORDAT), I/OPEN, FORS (Atos), DIALOG TOTAL, DTM / KIAS, Siline, IFAI / DIAFERT, MAN/X, FRIDA Command, IFAX Open, Compu Orga, JDE5 (J.D. Edwards), COMET, Bäurer b2 and much more
 - Adoption of released orders and operations
- Transfer of operation confirmations/uploads
 - Confirmation/upload of the collected actual times and quantities referred to transactions/ time tickets
- Monitoring
 - Comfortable monitoring and logging functions for the data transfer between HYDRA and the external systems

2 Enhancement for additional information from ERP

Use options

The function package additional information from ERP extends the ERP systems interface by additional data that are required for information or application purposes in the production.

Implementation notes

Use the additional information extension from ERP if:

- you use the ERP systems interface and
- your scope of data is not enough to map your requirements.

Integration

The function package integrates fully into the ERP system's interface.

Scope of functions

- Transfer of additional data
 - Extension of additional information from ERP
 - Adoption of component data, production resources and tools and document links to the operation
 - Adoption of long texts about the order header and operation
 - Adoption of customer-specific fields referring to the order header and to the operation

3 Setup of Data Record Structure

The data are transferred in the following structure. Within this structure the value of the SEGNAM field precisely defines the set-up of the user data structure in the SDATA field.

Field name	Type	Length	Designation	Data field and meaning
SEGNAM*	Char	30	Segment name	This field is occupied by the writing system with the respective segment name. This precisely defines the set-up of the data record (SDATA field). Example: HY72_AU_HD_001
MANDT*	Char	3	Client	Reserved; fixed: '000'
DOCNUM*	Char	16	IDOC number	Serial number for the IDOCs Reserved: fixed '0000000000000000'
SEGNUM*	Char	6	Segment number	Reserved: fixed '000000'
PSEGNUM	Char	6	Parent segment number	Reserved; fixed: '000000'
HLEVEL	Char	2	Hierarchy level	Reserved; fixed: '00'
SDATA	Char	1000	User data	This field contains the user data. The structure of this field is defined by the SEGNAM field.

4 Data type definitions

	→	
Type	Description	
CHAR x	The information is left-aligned for the data type CHAR; unnecessary places are filled with blanks (blanks - (U+0020)). If a field is not used, fill it in full length with blanks. Example: "ABCD "	
NUM x	Numeric field of the length x without sign. The NUMC data type only supports digits (ASCII characters 30 hex to 39 hex and/or U+0030 – U+0039). Numbers are right-aligned; unnecessary places are filled with zeros (U+0030). If a field is not used, fill it in full length with zeros. Example: "00000002"	
DEC x.y QUAN x.y	Numeric field of the length x and y decimal places. A data field in HYDRA format is preceded by a sign ("+" or "-") and includes a decimal point. Enter zeros to fill empty places. If a field is not used, fill it in full length with zeros (U+0030) including sign and decimal separator. e.g. DEC 13,3: -1234567890,123 → -1234567890.123 234567890,3 → +0234567890.300 Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. For example: a QUAN 13.3 field results in an external length of CHAR15.	
DATE	Dates must be transferred in the HYDRA format MM/DD/YYYY. Populate unused date fields with blanks (U+0020; zero(s) (U+0030) not accepted).	
TIME	Times must be transferred in the HYDRA format seconds after midnight (0 - 86400).	



For all alphanumeric fields, HYDRA does not support specific special characters. These characters are: "\" (backslash - U+005B), "|" (pipe - U+007C), „ “ “ (double quote - U+0022), and „ ’ “ (single quote - U+0027). You cannot enter these characters using the shop floor terminals;

the terminals and the MOC do not support these characters.



The character " ; " (semicolon - U+003B) is used as separator for data collection. You must not use this character in key fields (e.g. order, batch number, personnel number, etc.).



The character " % " (percent - U+0025) is used as placeholder/wildcard character for database queries. For this reason, you should avoid using this character as it might falsify results.



In general, you must not use special characters ranging from U+0000 to U+001F. Exception: U+000A and U+000D as end-of-line characters.



The file must not include [Byte Order Mark \(BOM\)](#).

In general, HYDRA always expects a contiguous data structure. Consequently, you have to populate unused data fields with such default values that comply with the applicable conventions. This also applies to fields that are not required at the end of a data structure. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) und 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, you can also transfer files in the file format that was used until MW 2.x.

5 Key Fields / Supported Characters

General notes

In all alphanumeric fields, HYDRA does not support specific special characters. The following characters are not supported: "%", "ß", "*", "\", "/", "|", "_", "?". Reason: You cannot enter these characters on the shop floor terminals or the clients do not support these characters.

You must not use the characters ";" (semicolon), "," (comma) and "'" (apostrophe) because they are often interpreted as comment characters or separators and can lead to unwanted results.

Workplace/machine numbers (resources of type "MNR")

Workplace/machine numbers and numbers of capacity/machine groups are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 8, left-aligned.

When you create or copy a workplace, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30_{hex} - 39_{hex}](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41_{hex} - 5A_{hex}](#))
- "-" ([US-ASCII 2D_{hex}](#))

Lower case letters are automatically converted to upper case letters when a new workplace is created.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

The entry "SYSTEM" as workplace/machine number is reserved for HYDRA and may not be used.

It is possible to overwrite the valid characters for the workplace/machine numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

Field	Value
Name	INPUT
Section	PATTERN
Key	MNR
Value	^(?!SYSTEM) ([A-Z0-9(){}~^#+!\$. _%-]+)\$
Active	☒

Minimum requirement: b_mnr.dll version 8.1.1.102

Resource numbers (resources of type <> "MNR")

Resource numbers are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 20; left-aligned.

When you create or copy a resource, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30_{hex} - 39_{hex}](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41_{hex} - 5A_{hex}](#))
- Umlauts "Ä", "Ö", "Ü" ([Extended ASCII C4_{hex}, D6_{hex}, DC_{hex}](#))
- "-" ([US-ASCII 2D_{hex}](#))

You may not use umlauts or special characters (e.g. "%", "ß", "*", "\", "/", "|", "_", "?") because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

Lower case letters are automatically converted to upper case letters when a new resource is created.

It is possible to overwrite the valid characters for the resource numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

Field	Value
Name	INPUT
Section	PATTERN
Key	RES
Value	^([0-9A-ZÄÖÜ(){}~^#+!\$. _%-]+)\$
Active	☒

Minimum requirement: b_res.dll version 8.1.1.117

HYDRA order number

There are some differences with respect to the order number in the HYDRA data model and the interface.

Order number

The order number (field AUNR) contains the actual order number as it is known in the ERP system and transferred to HYDRA. The order number is specified in the HYDRA basic settings; by default, this number has a length of 8 characters.

Operation number

The operation number (field AGNR) clearly identifies a defined process step of an order. The operation number is specified in the HYDRA basic settings; by default, this number has a length of 4 characters.

MES order number

The MES order number (field ANR) combines the order and the operation number and sometimes also the sequence number from the ERP system (if licensed). Its length therefore results from the total of the separate number lengths.

The total length must not exceed 25 digits. If DOS terminals are used, the total length must not exceed 16 digits.

Note the following for the order or operation number:

- Preferably only use the numbers "0" to "9" ([US-ASCII 30_{hex}](#) - [39_{hex}](#)).
- If you use letters, only the characters "A" - "Z" (upper case - [US-ASCII 41_{hex}](#) - [5A_{hex}](#)) and "-" ([US-ASCII 2D_{hex}](#)) are allowed. Do not use lower case letters.
- You may not use blanks in the numbers. The order or operation numbers must have the specified number of digits with the characters "0" to "9" or "A" to "Z". If required, you must prefix the numbers by leading zeros ("0").
- HYDRA does not support any umlauts, blank or special characters (see section *General notes*) for the order or operation number because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

6 Order Data ERP --> HYDRA

Data transfer process

Data of the ERP system is provided in a file that must be available in a defined directory of the HYDRA server. The configuration is made in the application ["Logical Systems"](#). You can combine the data transferred from the ERP system to HYDRA in one file to create/change data or to delete data. HYDRA uses the segment name of a data record (one line in the file) to identify if the respective data record must be added/updated (suffix "_A" for segment name) or if it must be deleted (suffix "_D" for segment name).

HYDRA processes the contents of the file sequentially top down. The higher a data record is placed in the file, the earlier it will be processed in HYDRA. This ensures that the file can be created in the chronologically correct order in the ERP system.



HYDRA 8 still supports but no longer improves the HYDRA 7 option ("compatibility mode") of transferring data to be created/changed and data to be deleted in separate files using different file extensions.

Create/change data

The interface transferring production orders bundles all data relevant to HYDRA in an IDOC and sends this data to HYDRA using the message function (= file extension) "DAT".

Make sure to transfer the data in a specific order, when transferring order data from the ERP system to HYDRA for the first time. This means that the order data described in the following sections must be transferred to HYDRA in the specified sequence/hierarchy. Within a transfer level, you are free to specify any sequence, e.g. you can transfer operation 0200 before operation 0100.

By default, you are not allowed to change operations with the status "running", "finished" or "deleted".

In general, HYDRA always expects a consistent data structure. This means: If a data field is not used, the predefined default value is assigned.



An operation in status "finished" is set to status "interrupted" when the operation is transferred again (update), if the following conditions are fulfilled:

- The operation has been deleted via ERP interface.
- The operation status set at the time of "delete": For this status, the action "delete" was configured in the order status configuration with "E" or "X".
- The operation is still in the online data set when transferred again (update).

If one of the conditions is not fulfilled, then an operation in status "finished" is not set to status "interrupted" when transferred again (update). The above also applies if the operation has been logged off via posting functionality (e.g. operation logoff).

Delete data

If entire orders are deleted in the ERP system, an IDOC is generated and sent to HYDRA using the message function (= file extension) "DAT". The data to be deleted must be transferred in the following order:

- Long texts of the operation
- Documents
- Production resources and tools
- Components
- Operation

When the last operation of an order is deleted via the interface , the order header is also deleted automatically¹.

If the ERP system transfers the deletion of an order header, all operations and the included data are deleted.

For deletion, it is sufficient to just indicate the respective key fields in the segment.

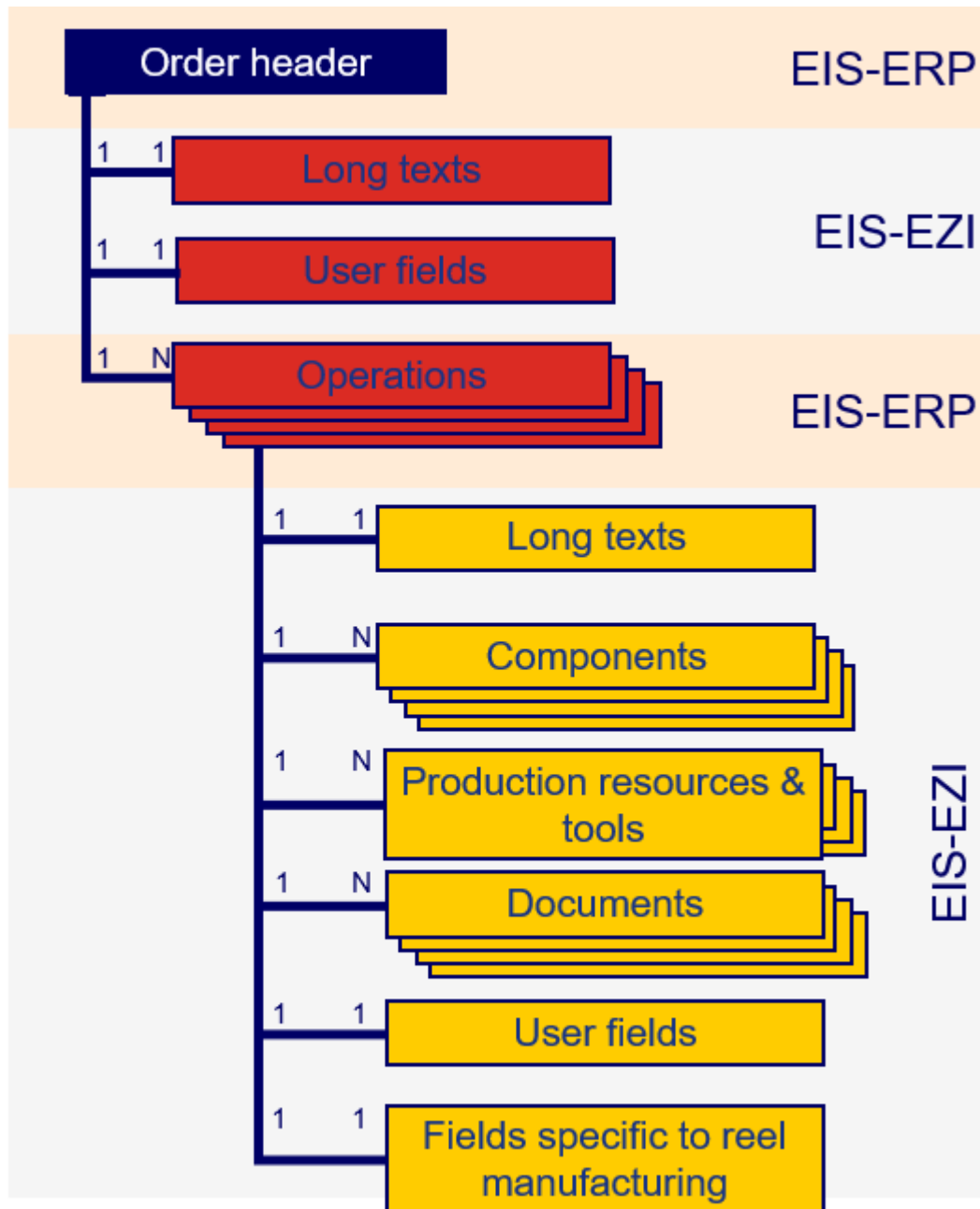
You cannot delete operations assigned to the operation statuses *Running*, *Interrupted*, *Finished* or *Deleted*.

You can neither delete operations that are split or part of a merged operation.

¹ If the last operation of an order is deleted via the MOC, the order header is not automatically deleted.

Data record definition

Order-related data is transferred for each production order in a multi-level IDOC. This IDOC structure is as follows:



The structures that belong to the individual segments are presented in the following. The individual columns have the following meaning:

Column	Description
Field	Name of the field

Column	Description
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be populated. M Mandatory field which must be populated with a valid value. ML Mandatory field if the HYDRA Shop Floor Scheduling is used (HLS). MM Mandatory field if the HYDRA Material and Production Logistics (MPL and/or MPL/RF) is used. K Field may stay empty (optional field). SA Mandatory field if the Arburg control system (SCS-ALS) is in use; otherwise optional field.
T(type)	Data type of the field
L(length)	Field length For fields of data type DEC: Total number of digits without decimal separator and algebraic sign.
D(decimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Field description and/or comments on the field.

Segments to create/modify data

The following specifications arise for the IDOC:

**Message type /
file name:**

HY72PPS

**Message
file extensions:**

functions/ DAT²

Segments:

```

HY72_AU_HD_001_A (order header)
├ HY72_AU_INFO_AI_001_A (long texts)
├ HY72_AU_USRFLD_001_A (user fields)
├ HY72_AG_HD_001_A (operation data – part 1)
│   ├── HY72_AG_HD_002_A (operation data – part 2)
│   ├── HY72_AG_KOMPL_001_A (component list)
│   │   └ HY72_AG_KOMPL_USRFLD_001_A (comp. user fields)
│   ├── HY72_AG_FHM_001_A (PRT / resources)
│   ├── HY72_AG_DOC_001_A (documents)
│   ├── HY72_AG_INFO_AI_001_A (long texts)
│   ├── HY72_AG_USRFLD_001_A (user fields)
│   └ HY72_AG_RF_001_A (MPL for data specific to coil-based
manufacturing)
└ HY72_FERTVAR_001_A (production variants)

```

² HYDRA inbound processing discards the file extension "DAT" (importing the file and storage in MLE inbound transactions). The MLE inbound transactions display a data record of such a file without message function.

Segments to delete data

The following specifications arise for the IDOC:

**Message type /
file name:**

HY72PPS

**Message
file extensions:**

functions/ DAT

Segments:

HY72_AU_HD_001_D (order header)
 ↳ HY72_AU_INFO_AI_001_D (long texts)
 ↳ HY72_AG_HD_001_D (operation data)
 ↳ HY72_AG_KOMPL_001_D (component list)
 ↳ HY72_AG_FHM_001_D (PRT / resources)
 ↳ HY72_AG_DOC_001_D (documents)
 ↳ HY72_AG_INFO_AI_001_D (long texts)

Order header

Segment name:	HY72_AU_HD_001_A HY72_AU_HD_001_D
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The order header includes the header data for the production order.

Field	V	T	L	D	Description	From ³	To ⁴
AUNR	S	CHAR	40		Order number	1	40
AUART	M	CHAR	5		Order type HYDRA order type (see note)	41	45
ATK	M	CHAR	40		Article; alphabetic characters in CAPITAL LETTERS	46	85
ATKBEZ	K	CHAR	40		Article designation	86	125
KDBEZ	K	CHAR	40		Customer name	126	165
KDAUNR	K	CHAR	25		Sales order Note: Using the MOC, you can only edit up to 20 digits/characters.	166	190
KDAUPOS	K	CHAR	15		Sales order item	191	205
EXTPRIO	K	CHAR	1		Priority	206	206

³ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

⁴ See footnote of column "From"

Field	V	T	L	D	Description	Fro m ³	To ⁴
AUIDX	K	DEC	4	2	Order index should be 0 (transfer: "+00.00") Please note: Make sure to add a sign and a decimal point to the length specified here. This applies to all following fields of the DEC data type	207	212
SGE:B	M	CHAR	3		Base quantity unit = quantity unit of the order header This quantity unit will be used for operations.	213	215
SGR:GUTB	M	DEC	13	3	Target quantity (base quantity unit)	216	230
SGR:AUSB	K	DEC	13	3	Target scrap quantity (base quantity unit)	231	245
MATTYP	K	CHAR	10		Material type of the article If HYDRA MPL is used, enter a valid material type configured in HYDRA. otherwise: "SYSTEM"	246	255
FILLER	K	CHAR	10		reserved; should be empty	256	265
CNR	K	CHAR	20		Batch number	266	285
PCNR	K	CHAR	20		Inspection order/inspection batch number	286	305
PPKTTYP	K	CHAR	1		Sample type	306	306
DATFB	ML	DATE	10		Earliest start (date)	307	316
ZEIFB	ML	TIME	5		Earliest start (time)	317	321
DATSE	ML	DATE	10		Latest end (date)	322	331
ZEISE	ML	TIME	5		Latest end (time)	332	336
DATTERMB	K	DATE	10		Scheduled start (date)	If scheduling is run outside HYDRA, the scheduled dates of the order (header) should be transferred. Note: For the processing, these dates are only used for information purposes in HYDRA. If scheduling is performed in HYDRA, these fields are overwritten.	346
ZEITERMB	K	TIME	5		Scheduled start (time)		351
DATERME	K	DATE	10		Scheduled end (date)		361
ZEITERME	K	TIME	5		Scheduled end (time)		366
TERMART	K	CHAR	1		Scheduling type; mandatory field if scheduling is to be made in HYDRA. (V=forward scheduling, R=backward scheduling)	367	367
REDSTRAT	K	CHAR	2		Reduction strategy, according to configuration.	368	369
AUGRP	K	CHAR	4		Order group	370	373
DISP	K	CHAR	10		MRP controller	374	383
PRJNR	K	CHAR	25		Project number	384	408
PLANAUNR	K	CHAR	25		Planned order	409	433
KTR	K	CHAR	25		Cost object	434	458
APNR	K	CHAR	40		Work plan	459	498
APVER	K	CHAR	12		Work plan version	499	510
SLVER	K	CHAR	12		BOM version	511	522
KLKK:MNR	K	DEC	13	3	Calc. costs - machines	523	537
KLKK:L	K	DEC	13	3	Calc. costs - payroll	538	552

Field	V	T	L	D	Description	Fro m ³	To ⁴
KLKK:MAT	K	DEC	13	3	Calc. costs - material	553	567
KLKK:SONST	K	DEC	13	3	Calc. costs - other	568	582
MATWERT:GUT	K	DEC	13	3	Material value	583	597
MATWERT:AUS	K	DEC	13	3	Scrap value	598	612
ANR.KBN:LBEZID	K	CHAR	15		eKANBAN control cycle This field is available as of the MLE variant HY72PPS_018	613	627
ATKIDX	K	CHAR	50		Order information → Order header (AK) → Drawing issue number This field is available as of BDE82 and MLE version HY72PPS_023.	628	677



In HYDRA the order type controls the global behavior of an order. The entered order type must exist in HYDRA and be entirely configured.

Long texts of the order header

Segment name:	HY72_AU_INFO_AI_001_A HY72_AU_INFO_AI_001_D
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Use the following structure to transfer text fields as additional information to the order(header) to HYDRA. The MOC will then show this additional information. Each data record contains one page of a maximum of 10 lines and 80 characters of text information each.

The long text of the order header is shown in the MOC application "edit orders".

Field	V	T	L	D	Description
KEY	S	CHAR	40		Order number (AUNR)
TYP	S	CHAR	2		Record type; by default: "AI"
SUBKEY:1	S	NUM	8		Reserved; assign "00000000" by default.
SUBKEY:2	M	NUM	8		Consecutive numbering within the key starting at "00000001".
INFO:BEZ	K	CHAR	20		Short text; if empty, the first 20 digits/characters of Info text 1 will be used.
INFO:1	K	CHAR	80		Info text 1
INFO:2	K	CHAR	80		Info text 2
INFO:3	K	CHAR	80		Info text 3
INFO:4	K	CHAR	80		Info text 4
INFO:5	K	CHAR	80		Info text 5
INFO:6	K	CHAR	80		Info text 6

Field	V	T	L	D	Description
INFO:7	K	CHAR	80		Info text 7
INFO:8	K	CHAR	80		Info text 8
INFO:9	K	CHAR	80		Info text 9
INFO:10	K	CHAR	80		Info text 10

User fields of the order header

Segment name:	HY72_AU_USRFLD_001_A
---------------	----------------------

Apart from the fields that are available by default, you can use user fields to store further customer-specific information to HYDRA. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the order(header).

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

Field	V	T	L	D	Description	From ⁵	To ⁶
AUNR	S	CHAR	40		Order number	1	40
USRFLD	S	CHAR	8		User field key. The user field key must be configured in the system.	41	48
FU:1	K	DATE	10		User field 1	49	58
FU:2	K	DATE	10		User field 2	59	68
FU:3	K	DATE	10		User field 3	69	78
FU:4	K	DATE	10		User field 4	79	88
FU:5	K	DATE	10		User field 5	89	98
FU:6	K	DATE	10		User field 6	99	108

⁵ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

⁶ See footnote of column "From"

Field	V	T	L	D	Description	From ⁵	To ⁶
FU:7	K	NUM	8		User field 7	109	116
FU:8	K	NUM	8		User field 8	117	124
FU:9	K	NUM	8		User field 9	125	132
FU:10	K	NUM	8		User field 10	133	140
FU:11	K	NUM	8		User field 11	141	148
FU:12	K	NUM	8		User field 12	149	156
FU:13	K	NUM	8		User field 13	157	164
FU:14	K	NUM	8		User field 14	165	172
FU:15	K	NUM	8		User field 15	173	180
FU:16	K	NUM	8		User field 16	181	188
FU:17	K	NUM	8		User field 17	189	196
FU:18	K	NUM	8		User field 18	197	204
FU:19	K	NUM	8		User field 19	205	212
FU:20	K	NUM	8		User field 20	213	220
FU:21	K	NUM	8		User field 21	221	228
FU:22	K	NUM	8		User field 22	229	236
FU:23	K	DEC	13	3	User field 23	237	251
FU:24	K	DEC	13	3	User field 24	252	266
FU:25	K	DEC	13	3	User field 25	267	281
FU:26	K	DEC	13	3	User field 26	282	296
FU:27	K	DEC	13	3	User field 27	297	311
FU:28	K	DEC	13	3	User field 28	312	326
FU:29	K	CHAR	1		User field 29	327	327
FU:30	K	CHAR	1		User field 30	328	328
FU:31	K	CHAR	1		User field 31	329	329
FU:32	K	CHAR	1		User field 32	330	330
FU:33	K	CHAR	1		User field 33	331	331
FU:34	K	CHAR	1		User field 34	332	332
FU:35	K	CHAR	1		User field 35	333	333
FU:36	K	CHAR	1		User field 36	334	334
FU:37	K	CHAR	1		User field 37	335	335
FU:38	K	CHAR	1		User field 38	336	336
FU:39	K	CHAR	1		User field 39	337	337
FU:40	K	CHAR	1		User field 40	338	338
FU:41	K	CHAR	1		User field 41	339	339
FU:42	K	CHAR	1		User field 42	340	340

Field	V	T	L	D	Description	From ⁵	To ⁶
FU:43	K	CHAR	1		User field 43	341	341
FU:44	K	CHAR	1		User field 44	342	342
FU:45	K	CHAR	10		User field 45	343	352
FU:46	K	CHAR	10		User field 46	353	362
FU:47	K	CHAR	10		User field 47	363	372
FU:48	K	CHAR	10		User field 48	373	382
FU:49	K	CHAR	10		User field 49	383	392
FU:50	K	CHAR	10		User field 50	393	402
FU:51	K	CHAR	20		User field 51	403	422
FU:52	K	CHAR	20		User field 52	423	442
FU:53	K	CHAR	20		User field 53	443	462
FU:54	K	CHAR	20		User field 54	463	482
FU:55	K	CHAR	20		User field 55	483	502
FU:56	K	CHAR	20		User field 56	503	522
FU:57	K	CHAR	20		User field 57	523	542
FU:58	K	CHAR	20		User field 58	543	562
FU:59	K	CHAR	20		User field 59	563	582
FU:60	K	CHAR	20		User field 60	583	602
FU:61	K	CHAR	20		User field 61	603	622
FU:62	K	CHAR	20		User field 62	623	642
FU:63	K	CHAR	20		User field 63	643	662
FU:64	K	CHAR	20		User field 64	663	682
FU:65	K	CHAR	40		User field 65	683	722
FU:66	K	CHAR	40		User field 66	723	762

Note

This data structure is only available with the respective license.

Order sequences

Segment name:	HY72_AFOLG_001_A HY72_AFOLG_001_D
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Only use this segment to transfer data if sequences are in use. This specification is made as part of the customer project.

Field	V	T	L	D	Description
AUNR	S	CHAR	40		Order number
AFOLG	S	CHAR	6		Number of the sequence Number of the sequence. Length as configured in the HYDRA basic settings, left-aligned including leading zeros. Sequence type "S": must be 0 Sequence type "P", "A": sequence number
FOLGART	S	CHAR	1		Sequence type: S = master sequence P = parallel sequence A = alternative sequence
AKTIV	M	CHAR	1		Active; with alternative sequences J/N; otherwise always J
BEZK	K	CHAR	40		Short text/description of the sequence: comment field
VER	K	CHAR	12		Version; comment field; not processed in HYDRA
BZGFOLG	M	CHAR	10		Reference sequence. Number of the reference sequence this sequence refers to. Length as configured in the HYDRA basic settings, left-aligned including leading zeros. Sequence type "S": leave empty Sequence type "A", "P": must always be 0.
ANRA	M	CHAR	40		Branch operation of the reference sequence; combined HYDRA order number Sequence type "S": leave empty
ANRR	M	CHAR	40		Return OP of the reference sequence; combined HYDRA order number Sequence type "S": leave empty

Operations – part 1

Segment name:	HY72_AG_HD_001_A HY72_AG_HD_001_D
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The operation segment includes the operation-based specifications for production and relevant data.

Field	V	T	L	D	Description	From	To ⁸
ANR	S	CHAR	40		Combined order/OP number	1	40
AUART	K	CHAR	5		Order type according to the configuration in HYDRA. The order type must be the same as in the order header and be configured in HYDRA. The attempt of transferring an order type that deviates from the order header is ignored. If this field remains empty, the order type specified in the header is used .	41	45
AGBEZ	M	CHAR	40		Operation designation/name.	46	85
ATK	K	CHAR	40		Article/item (number); HYDRA only displays up to 25 digits/characters. Alphabetic characters in CAPITAL LETTERS If no item/article number is transferred when creating an operation (blank), the article number of the order header will be used for the operation. If no item/article number is transferred when an operation is changed, the article number of the order header will be used.	86	125
ATKBEZ	K	CHAR	40		Item/article designation/name; HYDRA only displays up to 25 digits. If no item/article designation/name is transferred when an operation is created (blank), the article designation/name of the order header will be used for the operation (redundant). It is recommended to leave this field blank. If no item/article designation/name is transferred when changing an operation, the item/article designation/name of the order header is used. Please note: the item/article designation/name of the order header can only be edited in the MOC; it is automatically used for all operations of the order when saving.	126	165
MATTYP	MM	CHAR	10		Material type of the item/article; when HYDRA MPL is used, according to the material type configuration If no material type is transferred when creating an operation, the material type of the order header is used. If no material type is transferred when changing an operation, the material type of the order header is used.	166	175
FILLER	K	CHAR	10		Reserved; must be blank.	176	185
EXTPRIO	M	CHAR	1		Priority (0 - 9; 9 = priority high) If the priority check is enabled for the order type, the priority is always transferred from the order header to the operations. Any deviating priorities of the operation are ignored.	186	186
MNR	M	CHAR	8		Planned workplace At least one of the fields MNR and/or MGRP must be transferred. When transferring the workplace, HYDRA identifies the workplace's group according to the configuration to avoid inconsistent data. The transferred group is then ignored.	187	194

⁷ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

⁸ See footnote of column "From"

Field	V	T	L	D	Description	From	To ⁸															
MGRP	M	CHAR	8		Planned group and/or group of the workplace At least one of the fields MNR and/or MGRP must be transferred. If no machine group is transferred, the system automatically identifies the machine group using the transferred workplace.	195	202															
OPT:PLAN	M	CHAR	1		Planned M - operation is planned (in detail) for the workplace G - operation is in the pool of (groups) (MNR empty) Depending on the workplace (MNR field), the indicator is set internally according to the following logic: <table><tr><th>Workplace MNR</th><th>Planned OPT:PLAN</th><th>Result</th></tr><tr><td>blank</td><td>blank/ "M"/"G" →</td><td>OPT:PLAN is set to "G"</td></tr><tr><td>not blank to "M"</td><td>blank</td><td>→ OPT:PLAN is set to "M"</td></tr><tr><td>not blank remains at "M"</td><td>"M"</td><td>→ OPT:PLAN remains at "M"</td></tr><tr><td>not blank remains at "G"</td><td>"G"</td><td>→ OPT:PLAN remains at "G"</td></tr></table>	Workplace MNR	Planned OPT:PLAN	Result	blank	blank/ "M"/"G" →	OPT:PLAN is set to "G"	not blank to "M"	blank	→ OPT:PLAN is set to "M"	not blank remains at "M"	"M"	→ OPT:PLAN remains at "M"	not blank remains at "G"	"G"	→ OPT:PLAN remains at "G"	203	203
Workplace MNR	Planned OPT:PLAN	Result																				
blank	blank/ "M"/"G" →	OPT:PLAN is set to "G"																				
not blank to "M"	blank	→ OPT:PLAN is set to "M"																				
not blank remains at "M"	"M"	→ OPT:PLAN remains at "M"																				
not blank remains at "G"	"G"	→ OPT:PLAN remains at "G"																				
RES:WNR	K	CHAR	40		(Main) tool; see notes (see below)!	204	243															
RES:DNC	K	CHAR	40		NC program; see notes (see below)!	244	283															
RES:EMAT	K	CHAR	40		(Main) input material; see notes (see below)!	284	323															
COLOR	K	CHAR	20		Color of the material	324	343															
KST	K	CHAR	8		Cost center	344	351															
KART	K	CHAR	10		Cost type	352	361															
ASTUFE	K	CHAR	1		Authorization level to log in and off the OP (lowest authorization = 1)	362	362															
RMNR	K	CHAR	10		Confirmation/upload number	363	372															
DATTERM B	K	DATE	10		Scheduled start (date)	If scheduling is run outside HYDRA, the scheduled dates of the operations must be transferred from the ERP system.	373	382														
ZEITERM B	K	TIME	5		Scheduled start (time)		383	387														
DATTERM E	K	DATE	10		Scheduled end (date)		388	397														
ZEITERM E	K	TIME	5		Scheduled end (time)	If scheduling is performed in HYDRA, these fields are overwritten. In HYDRA Shop Floor Scheduling (HLS), the operations are displayed in the pool of groups of the graphic planning board for the scheduled start date.	398	402														
DATF B	K	DATE	10		Earliest start (date)				403	412												
ZEIF B	K	TIME	5		Earliest start (time)				413	417												
DATSB	K	DATE	10		Latest start (date)	If scheduling is performed in HYDRA, these fields are overwritten. The basic dates result from forward and backward scheduling in HYDRA.	418	427														
ZEISB	K	TIME	5		Latest start (time)		428	432														
DATFE	K	DATE	10		Earliest end (date)		433	442														
ZEIFE	K	TIME	5		Earliest end (time)		443	447														
DATSE	K	DATE	10		Latest end (date)		448	457														

Field	V	T	L	D	Description	From	To ⁸
ZEISE	K	TIME	5		Latest end (time)	458	462
DATB	K	DATE	10		Planned start (date)	463	472
ZEIB	K	TIME	5		Planned start (time)	473	477
DATE	K	DATE	10		Planned end (date)	478	487
ZEIE	K	TIME	5		Planned end (time)	488	492
					If planning is performed in HYDRA, these fields are overwritten. Note: The planned dates must include valid values if the operation is directly planned for a workplace (OPT:PLAN=M).		
SGR:GUTB	K	DEC	13	3	Target quantity (base quantity unit)	493	507
SGR:GUTP	M	DEC	13	3	Target quantity (primary quantity unit)	508	522
SGR:GUTS	K	DEC	13	3	Target quantity (secondary quantity unit)	523	537
SGR:GUTT	K	DEC	13	3	Target quantity (tertiary quantity unit)	538	552
SGR:AUSB	K	DEC	13	3	Target scrap quantity (base quantity unit)	553	567
SGR:AUSP	K	DEC	13	3	Target scrap quantity (primary quantity unit)	568	582
SGR:AUSS	K	DEC	13	3	Target scrap quantity (secondary quantity unit)	583	597
SGR:AUST	K	DEC	13	3	Target scrap quantity (tertiary quantity unit)	598	612
SGE:B	K	CHAR	3		Base quantity unit If no base quantity unit is transferred when creating an operation, the unit of the order header is used, if applicable. If no base quantity unit is transferred when changing an operation, the unit of the order header is used, if applicable.	613	615
SGE:P	M	CHAR	3		Primary input quantity unit	616	618
SGE:S	K	CHAR	3		Secondary input quantity unit	619	621
SGE:T	K	CHAR	3		Tertiary input quantity unit	622	624
WEIGMENGE	K	DEC	13	3	Minimum send-ahead quantity (primary quantity unit)	625	639
MENGEPROZ:UNTLL	K	DEC	13	3	Underdelivery in percent The value entered is an absolute percentage value of the target quantity (primary quantity unit). Example: - Target quantity of the operation: 120 pieces - Underdelivery: 84% The actual quantity must not fall below 101 items.	640	654
OPT:UNTLL	K	CHAR	1		Reaction to underdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; underdelivery not allowed. Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").	655	655

Field	V	T	L	D	Description	From	To ⁸
MENGEPROZ:UEBLI	K	DEC	13	3	Overdelivery in percent The value entered is an absolute percentage value of the target quantity (primary quantity unit). Example: - Target quantity of the operation: 120 pieces - Overdelivery: 168% The actual quantity must not exceed 201 items.	656	670
OPT:UEBLI	K	CHAR	1		Reaction to overdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; overdelivery not allowed. Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").	671	671
UMRFAKTP:Z	K	NUM	8		Numerator for the conversion of primary quantity to basic quantity Example: - Base quantity unit: square meter M2 - Primary quantity unit: piece ST - 1 piece = 2 square meters. In this case to be stored as - numerator (= UMRFAKTP:Z) 2 and - denominator (= UMRFAKTP:N) 1. MPL for coil-based manufacturing: the conversion is based on specific conversion factors (see the RF structure below); therefore 0 must be transferred for operations that are produced at machines for coil-based production or at coil cutting machines.	672	679
UMRFAKTP:N	K	NUM	8		Denominator for the conversion of primary quantity to basic quantity MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field	680	687
UMRFAKTS:Z	K	NUM	8		Numerator for the conversion of secondary quantity to basic quantity MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field	688	695
UMRFAKTS:N	K	NUM	8		Denominator for the conversion of secondary quantity to basic quantity MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field	696	703
UMRFAKTT:Z	K	NUM	8		Numerator for the conversion of tertiary quantity to basic quantity MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field	704	711
UMRFAKTT:N	K	NUM	8		Denominator for the conversion of tertiary quantity to basic quantity MPL for coil-based manufacturing: see the note in the UMRFAKTP:Z field	712	719
RUEZ	K	NUM	8		Setup time in seconds. If no setup time exists the value is explicitly to be set to 0.	720	727
RUEZ:ZUSCHL	K	NUM	8		Additional setup time in seconds. Should be set to 0 if not available.	728	735
BEARBZEI	K	NUM	8		Processing time in seconds. Should be set to 0 if not available.	736	743
PZ	K	NUM	8		Inspection time in seconds. Should be set to 0 if not available.	744	751

Field	V	T	L	D	Description	From	To ⁸
ABRZ	K	NUM	8		Teardown/retooling time in seconds. Should be set to 0 if not available.	752	759
LIZ	K	NUM	8		Delivery time in seconds At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should be set to 0 if it is not an external processing OP.	760	767
FREMDFERT	K	CHAR	1		External processing OP Y/N At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should in general be set to "N".	768	768
RLZ:EXPR	K	CHAR	6		Remaining run time (formula 1); Relevant if the HYDRA Shop Floor Scheduling module (HLS) and specific BDE applications are used. Generally, transfer "RLFZ" by default. Configure any deviating remaining run time formulas in the system's Formula management .	769	774
RLZ2:EXPR	K	CHAR	6		Remaining run time (formula 2); option (leave empty)	775	780
VLZ	K	NUM	8		Lead time in seconds. Should be set to 0 if not available.	781	788
LIEZ:MAX	K	NUM	8		Max. synchronization time in seconds. Should be set to 0 if not available.	789	796
WARTZ	K	NUM	8		Waiting time in seconds. Should be set to 0 if not available.	797	804
WARTZ:MIN	K	NUM	8		Minimum waiting time in seconds. Should be set to 0 if not available.	805	812
LIEZ	K	NUM	8		Idle period in seconds. Should be set to 0 if not available.	813	820
LART	K	CHAR	4		Wage type	821	824
AKKORD	K	CHAR	1		Piecework indicator/ premium	825	825
TE	K	DEC	13	3	Premium specification te in seconds per 1000 pieces. Should be set to 0 if not available.	826	840
TR	K	DEC	13	3	Premium specification tr in seconds. Should be set to 0 if not available.	841	855
TEB	K	DEC	13	3	Premium specification teb in seconds per 1000 pieces. Should be set to 0 if not available.	856	870
TRB	K	DEC	13	3	Premium specification trb in seconds. Should be set to 0 if not available.	871	885
VERARBCODE	M	CHAR	6		Processing code; "SYSTEM" by default. You can configure additional processing codes in the system.	886	891
OPT:ERF	M	CHAR	1		Recordable true/false	892	892
OPT:MULTIMNR	M	CHAR	1		Parallel production true/false	893	893
OPT:CNR	MM	CHAR	1		Batch management requirement true/false ("true" only relevant for MPL / TRT and/or AIP-MTR)	894	894
OPT:SNR	MM	CHAR	1		Serial numbers required (only relevant if serial numbers are used) "G" = Automatic generation of the serial number (only in combination with MPL as batch number) "S" = Automatic generation of the serial number using number range (only in combination with MPL as batch number) "E" = Manual entry of the serial number " " or "N" = serial numbers are not used For details on the implementation/configuration of serial numbers, see here .	895	895
SZY	K	NUM	8		Target cycle in seconds/ 1000; should be set; mandatory for MDE cycle monitoring	896	903
TLG	K	NUM	8		Partitioning; should be pre-populated with 1	904	911

Field	V	T	L	D	Description	From	To ⁸
IMPFAKT	K	DEC	13	3	Pulse factor; only integer values allowed! Should be pre-populated with 1 by default.	912	926
OPT:SPLIT	K	CHAR	1		May be split V/N V = Yes, operation may be split N = No, operation must not be split Please note: V only relevant for BDE-SSG, ADE-SPL, HLS-AGS, N must be transferred in the other cases.	927	927
MAXANZSPLIT	K	NUM	8		Max. no. of splits. (only relevant if OPT:SPLIT = V)	928	935
MBVERH:RUE	K	DEC	5	2	Machine-operator relation: setup/ PEP workforce requirements: setup	936	942
MBVERH:NORM	K	DEC	5	2	Machine-operator relation manufacturing/ PEP workforce requirements: manufacturing	943	949
QUAL:NORM	K	NUM	8		PEP: Qualification: manufacturing	950	957
QUAL:RUE	K	NUM	8		PEP: Qualification: setup	958	965



Fields RES:WNR, RES:DNC, RES:EMAT

The fields

- tool (RES:WNR)
- NC program (RES:DNC)
- input material (RES:EMAT)

are displayed in different evaluation dialogs. Note the following if you populate these fields via the ERP interface:

Only set the operation fields if NO material components are transferred using the structure HY72_AG_KOMPL_001 and NO production resources and tools are transferred using the structure HY72_AG_FHM_001.

If an existing operation is changed, all entries of the component list and of the production resources and tools will automatically be deleted and must then explicitly be transferred.

If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the production resources and tools to HYDRA (no "delta download").



Use of the HYDRA shop floor scheduling system (HLS) or of the graphic order sequencing (BDE-GAV)

If the HYDRA shop floor scheduling (HLS) or the graphic order sequencing (BDE-GAV) are used, you should consider the following issues for operations that are planned with these applications:

You may only transfer operations planned for groups (OPT:PLAN= G). In this case, the

planner explicitly carries out the detailed planning (scheduling for the workplace).

If you want to transfer and display operations as planned for a workplace, it is mandatory to populate the following fields:

MNR=<Workplace where the operation must be planned>

PLAN:OPT=M

DATB=<Planned start, date>

ZEIB=<Planned start, time>

DATE=<Planned end, date>

ZEIE=<Planned end, time>

Operations – part 2

Segment name:	HY72_AG_HD_002_A
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The segment for the operation includes further operation-based specifications for production and relevant data.

Field	V	T	L	D	Description	From ⁹	To ¹⁰
ANR	S	CHAR	40		Combined order/OP number	1	40
ATKIDX	K	CHAR	50		Order information → OP → Drawing issue number This field is available as of BDE82 and MLE version HY72PPS_023.	41	90

Long texts of the operation

Segment name:	HY72_AG_INFO_AI_001_A HY72_AG_INFO_AI_001_D
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Use the following structure to transfer text fields as additional information on the operation to HYDRA. The texts are then displayed in the MOC. Each data record contains one page of a maximum of 10 lines and 80 characters of text information each.

The long text of an operation is shown, e.g. in:

- the MOC application "edit operation"

⁹ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

¹⁰ See footnote of column "From"

- the MOC application "order information"
- on the AIP shop floor terminal

Field	V	T	L	D	Description
KEY	S	CHAR	40		Combined order/OP number
TYP	S	CHAR	2		Record type; by default: "AI"
SUBKEY:1	S	NUM	8		Reserved; "00000000" by default.
SUBKEY:2	M	NUM	8		Consecutive numbering within the key starting at "00000001".
INFO:BEZ	K	CHAR	20		Short text; only relevant for SUBKEY:2 = "00000001". If empty, the first 20 digits from Info text 1 will be used
INFO:1	K	CHAR	80		Info text 1
INFO:2	K	CHAR	80		Info text 2
INFO:3	K	CHAR	80		Info text 3
INFO:4	K	CHAR	80		Info text 4
INFO:5	K	CHAR	80		Info text 5
INFO:6	K	CHAR	80		Info text 6
INFO:7	K	CHAR	80		Info text 7
INFO:8	K	CHAR	80		Info text 8
INFO:9	K	CHAR	80		Info text 9
INFO:10	K	CHAR	80		Info text 10

Components – Component data

Segment name (OBSOLETE):	HY72_AG_KOMPL_001_A HY72_AG_KOMPL_001_D
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Segment name:	HY72_AG_KOMPL_002_A HY72_AG_KOMPL_002_D
Segment name in compatibility mode:	HY72_AG_KOMPL_002

Field	V	T	L	D	Description	From ¹¹	To ¹²
ANR	S	CHAR	40		Combined order/OP number	1	40
ATK	S	CHAR	40		Material number; alphabetic characters in CAPITAL LETTERS	41	80

¹¹ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

¹² See footnote of column "From"

Field	V	T	L	D	Description	From ¹¹	To ¹²
ATKBEZ	M	CHAR	40		Material designation/name	81	120
BEZ	K	CHAR	30		Comment 1	121	150
BEZ:2	K	CHAR	30		Comment 2	151	180
SLP	S	CHAR	10		BOM item MPL for coil-based manufacturing: item of the component in the layer structure. Each component must have a unique BOM item if several components are used in one operation. Two components must not have the same BOM item.	181	190
SLS	M	NUM	8		BOM level Material components with the BOM level > 1 will always be saved under the material type "I" = info component. If you log on input batches via material management (MPL/TRT), you can only log on components of BOM level 0.	191	198
ART	MM	CHAR	2		Material type: "M" (Consumption) material "T" Carrier material (only MPL-RF) "A" Waste component (only MPL-RF) "Z" Additional material (only MPL-RF) "I" Info component	199	200
MATTYP	MM	CHAR	10		MPL: Material type If HYDRA MPL is used, enter a valid material type configured in HYDRA. otherwise: "SYSTEM"	201	210
VERBR	MM	CHAR	1		MPL: Consumption type If not specified otherwise, assign "L" to this field. For components of the material type "I" assign "N" to the field. For discrete consumption, assign "D" to the field.	211	211
OPT:ERSB	MM	CHAR	1		MPL-RF: Replaceable – is it possible to use another material than the planned one for such a component? You may only use the same material type that matches the type of the material to be produced. You may only log unplanned material on to the machine, if the respective status is used. MPL: true/false otherwise: "N" by default	212	212
OPT:WZW	MM	CHAR	1		MPL: Change necessary; an input batch change for a batch of this material requires an output batch change: if ART = "T" or "Z" -> OPT:WZW must be "J" if ART = "I" or "A" -> OPT:WZW must be "N" if ART = "M" -> OPT:WZW: "J" or "N"	213	213
SGR:GUT	MM	DEC	13	3	MPL: Input quantity to produce 1 article in primary quantity unit of the operation.	214	228
SGE:GUT	MM	CHAR	3		MPL: Quantity unit of the input quantity	229	231
MENGEPROZ	K	DEC	13	3	Input quantity in percent Reserved; currently not used.	232	246

Field	V	T	L	D	Description	From ¹¹	To ¹²
UTL	K	DEC	13	3	Upper tolerance limit in percent; 3 decimal places Reserved; currently not used.	247	261
LTL	K	DEC	13	3	Lower tolerance limit in percent; 3 decimal places Reserved; currently not used.	262	276
EGR:GUT	K	DEC	13	3	Total quantity/required quantity: total quantity required for the OP, i.e. for the quantity to be produced (output quantity)	277	291
EGE:GUT	K	CHAR	3		Unit of the required quantity	292	294
SLS:M	K	NUMC	8		BOM level of the parent material Reserved; currently not used.	295	302
SLP:M	K	CHAR	10		BOM item of the parent material Reserved; currently not used.	303	312
MENGE:FIX	K	CHAR	1		Indicator: Fixed quantity Reserved; currently not used.	313	313
PPS:RETRO	K	CHAR	1		Indicator: Backflush (retrograde withdrawal) (in ERP) Reserved; currently not used.	314	314
MENGETOL	K	DEC	13	3	Tolerance in percent Please note: only in segment HY72_AG_KOMPL_002_A	315	329
MENGEABW	K	DEC	13	3	Deviation in percent Please note: only in segment HY72_AG_KOMPL_002_A	330	344
OPT:SLOS	K	CHAR	1		Flag for merged batch As of MPL82 / MLE version HY72PPS_019 Identification of the component as part of the collection of serial numbers for merging " " If no serial numbers are collected and if serial numbers are collected and merged: subordinated serial number "F" if serial numbers are collected and merged: leading serial number	345	345
VERB:ZAEHLER	K	CHAR	1		Indicator for automatic counter consumption (as of MPL82 / MLE version HY72PPS_019): "J" Automatic counter consumption enabled "N" Automatic counter consumption <i>disabled</i>	346	346

If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the production resources and tools to HYDRA (no "delta download").



Also see the notes in the "Operations" chapter.

During the data transfer to HYDRA, the first material transferred to HYDRA will be displayed in the operation.

Component - user fields

Segment name:	HY72_AG_KOMPL_USRFLD_001_A
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Use user fields to store further customer-specific information to HYDRA in addition to the fields that are available by default. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the component.

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

Field	V	T	L	D	Description	From ¹³	To ¹⁴
ANR	S	CHAR	40		Combined order/OP number	1	40
ATK	S	CHAR	40		Material number; alphabetic characters in CAPITAL LETTERS	41	80
SLP	S	CHAR	10		BOM item MPL for coil-based manufacturing: item of the component in the layer structure.	81	90
USRFLD	S	CHAR	8		User field key. The user field key must be configured in the system.	91	98
FU:1	K	DATE	10		User field 1	99	108
FU:2	K	DATE	10		User field 2	109	118
FU:3	K	DATE	10		User field 3	119	128
FU:4	K	DATE	10		User field 4	129	138
FU:5	K	DATE	10		User field 5	139	148
FU:6	K	DATE	10		User field 6	149	158
FU:7	K	NUM	8		User field 7	159	166
FU:8	K	NUM	8		User field 8	167	174
FU:9	K	NUM	8		User field 9	175	182
FU:10	K	NUM	8		User field 10	183	190

¹³ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA application developers. It does not include the number of characters of the file itself but only of the field SDATA.

¹⁴ See footnote of column "From"

Field	V	T	L	D	Description	From 13	To ¹⁴
FU:11	K	NUM	8		User field 11	191	198
FU:12	K	NUM	8		User field 12	199	206
FU:13	K	NUM	8		User field 13	207	214
FU:14	K	NUM	8		User field 14	215	222
FU:15	K	NUM	8		User field 15	223	230
FU:16	K	NUM	8		User field 16	231	238
FU:17	K	NUM	8		User field 17	239	246
FU:18	K	NUM	8		User field 18	247	254
FU:19	K	NUM	8		User field 19	255	262
FU:20	K	NUM	8		User field 20	263	270
FU:21	K	NUM	8		User field 21	271	278
FU:22	K	NUM	8		User field 22	279	286
FU:23	K	DEC	13	3	User field 23	287	301
FU:24	K	DEC	13	3	User field 24	302	316
FU:25	K	DEC	13	3	User field 25	317	331
FU:26	K	DEC	13	3	User field 26	332	346
FU:27	K	DEC	13	3	User field 27	347	361
FU:28	K	DEC	13	3	User field 28	362	376
FU:29	K	CHAR	1		User field 29	377	377
FU:30	K	CHAR	1		User field 30	378	378
FU:31	K	CHAR	1		User field 31	379	379
FU:32	K	CHAR	1		User field 32	380	380
FU:33	K	CHAR	1		User field 33	381	381
FU:34	K	CHAR	1		User field 34	382	382
FU:35	K	CHAR	1		User field 35	383	383
FU:36	K	CHAR	1		User field 36	384	384
FU:37	K	CHAR	1		User field 37	385	385
FU:38	K	CHAR	1		User field 38	386	386
FU:39	K	CHAR	1		User field 39	387	387
FU:40	K	CHAR	1		User field 40	388	388
FU:41	K	CHAR	1		User field 41	389	389
FU:42	K	CHAR	1		User field 42	390	390
FU:43	K	CHAR	1		User field 43	391	391
FU:44	K	CHAR	1		User field 44	392	392
FU:45	K	CHAR	10		User field 45	393	402

Field	V	T	L	D	Description	From 13	To ¹⁴
FU:46	K	CHAR	10		User field 46	403	412
FU:47	K	CHAR	10		User field 47	413	422
FU:48	K	CHAR	10		User field 48	423	432
FU:49	K	CHAR	10		User field 49	433	442
FU:50	K	CHAR	10		User field 50	443	452
FU:51	K	CHAR	20		User field 51	453	472
FU:52	K	CHAR	20		User field 52	473	492
FU:53	K	CHAR	20		User field 53	493	512
FU:54	K	CHAR	20		User field 54	513	532
FU:55	K	CHAR	20		User field 55	533	552
FU:56	K	CHAR	20		User field 56	553	572
FU:57	K	CHAR	20		User field 57	573	592
FU:58	K	CHAR	20		User field 58	593	612
FU:59	K	CHAR	20		User field 59	613	632
FU:60	K	CHAR	20		User field 60	633	652
FU:61	K	CHAR	20		User field 61	653	672
FU:62	K	CHAR	20		User field 62	673	692
FU:63	K	CHAR	20		User field 63	693	712
FU:64	K	CHAR	20		User field 64	713	732
FU:65	K	CHAR	40		User field 65	733	772
FU:66	K	CHAR	40		User field 66	773	812

Production resources and tools

Segment name:	HY72_AG_FHM_001_A HY72_AG_FHM_001_D
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If you want to use production resources and tools as resources in HYDRA, you also have to use the HYDRA Tool and Resource Management (WRM) and/or of HYDRA DNC. If production resources and tools are only displayed in the BDE module, you do not have to use WRM or DNC.

Field	K	T	L	Description
ANR	S	CHAR	40	Combined order/OP number

Field	K	T	L		Description
RESTYP	S	CHAR	4		Resource type; possible values: DNC DNC program ENT removal device TEM temperature device VOR device WNR tool If you use the HYDRA tools and resources management (WRM), you can define additional resource types (idents) in the application's resource type configuration (Menu <i>WRM: Master data</i> > <i>Resource types</i>).
ATK	S	CHAR	40		Resource/ material number; alphabetic characters in CAPITAL LETTERS
ATKBEZ	K	CHAR	40		Designation
BEZ	K	CHAR	30		Comment 1
BEZ:2	K	CHAR	30		Comment 2
SGR:GUT	M	DEC	13	3	Input quantity Please note: The quantity of a production resource and tool that is to be checked in the HYDRA shop floor scheduling module (license: HLS-BSR) is generally assumed to be 1, if 0 has been transferred here.
SGE:GUT	K	CHAR	3		Quantity unit



If a material component changes (change of a material component) or if a production resource and tool is changed (change of a tool), the ERP system must transfer the changed operation as well as the complete material list and the complete list of the production resources and tools to HYDRA (no "delta download").

Also see the notes in the "Operations" chapter.

During the data transfer to HYDRA, the first production resource and tool transferred with RESTYP=DNC to HYDRA will be taken over in the operation and displayed as NC program.

Otherwise, the first production resource and tool transferred with RESTYP<>DOC to HYDRA will be taken over for the operation and displayed as tool. If production resources and tools of different resource types are transferred, the customer must ensure a useful sequence (e.g. first the prod. resources + tools of the resource type WNR and then the production resources and tools of another resource type).

Use the option "automatic creation" in the [Resource types](#) configuration to make sure that production resources and tools of a resource type are created automatically as resource, provided that they do not yet exist in the system (only relevant if the WRM product group (Tool and Resource Management) is in use).

Documents

Segment name:	HY72_AG_DOC_001_A HY72_AG_DOC_001_D
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Field	V	T	L	D	Description
ANR	S	CHAR	40		Combined order/OP number
ATK	S	CHAR	40		Document ID: unique key alphabetic characters in CAPITAL LETTERS
ATKBEZ	M	CHAR	40		Designation
BEZ	K	CHAR	30		Comment 1
BEZ:2	K	CHAR	30		Comment 2
PATH	M	CHAR	8		Refers to a path defined in the Path configuration .
FILE	M	CHAR	128		File name incl. file extension

Operation user fields

Segment name:	HY72_AG_USRFLD_001_A
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Use user fields to store further customer-specific information to HYDRA in addition to the fields that are available by default. Use this segment to transfer this data from the ERP system to HYDRA and to store this data in the operation.

The so-called user field key specifies the available user fields and their meaning. Each user field key describes a combination of user fields. The document [Configuration Userfields.pdf](#) describes how to configure the user field key.



To allow for a consistent data exchange between the ERP system and HYDRA, the customer must ensure that the user field keys are identical in both systems.

Field	V	T	L	D	Description	From	To
ANR	S	CHAR	40		Combined order/OP number	1	40
USRFLD	S	CHAR	8		User field key. The user field key must be configured in the system.	41	48
FU:1	K	DATE	10		User field 1	49	58
FU:2	K	DATE	10		User field 2	59	68
FU:3	K	DATE	10		User field 3	69	78
FU:4	K	DATE	10		User field 4	79	88
FU:5	K	DATE	10		User field 5	89	98
FU:6	K	DATE	10		User field 6	99	108
FU:7	K	NUM	8		User field 7	109	116

FU:8	K	NUM	8		User field 8	117	124
FU:9	K	NUM	8		User field 9	125	132
FU:10	K	NUM	8		User field 10	133	140
FU:11	K	NUM	8		User field 11	141	148
FU:12	K	NUM	8		User field 12	149	156
FU:13	K	NUM	8		User field 13	157	164
FU:14	K	NUM	8		User field 14	165	172
FU:15	K	NUM	8		User field 15	173	180
FU:16	K	NUM	8		User field 16	181	188
FU:17	K	NUM	8		User field 17	189	196
FU:18	K	NUM	8		User field 18	197	204
FU:19	K	NUM	8		User field 19	205	212
FU:20	K	NUM	8		User field 20	213	220
FU:21	K	NUM	8		User field 21	221	228
FU:22	K	NUM	8		User field 22	229	236
FU:23	K	DEC	13	3	User field 23	237	251
FU:24	K	DEC	13	3	User field 24	252	266
FU:25	K	DEC	13	3	User field 25	267	281
FU:26	K	DEC	13	3	User field 26	282	296
FU:27	K	DEC	13	3	User field 27	297	311
FU:28	K	DEC	13	3	User field 28	312	326
FU:29	K	CHAR	1		User field 29	327	327
FU:30	K	CHAR	1		User field 30	328	328
FU:31	K	CHAR	1		User field 31	329	329
FU:32	K	CHAR	1		User field 32	330	330
FU:33	K	CHAR	1		User field 33	331	331
FU:34	K	CHAR	1		User field 34	332	332
FU:35	K	CHAR	1		User field 35	333	333
FU:36	K	CHAR	1		User field 36	334	334
FU:37	K	CHAR	1		User field 37	335	335
FU:38	K	CHAR	1		User field 38	336	336
FU:39	K	CHAR	1		User field 39	337	337
FU:40	K	CHAR	1		User field 40	338	338
FU:41	K	CHAR	1		User field 41	339	339
FU:42	K	CHAR	1		User field 42	340	340
FU:43	K	CHAR	1		User field 43	341	341
FU:44	K	CHAR	1		User field 44	342	342

FU:45	K	CHAR	10		User field 45	343	352
FU:46	K	CHAR	10		User field 46	353	362
FU:47	K	CHAR	10		User field 47	363	372
FU:48	K	CHAR	10		User field 48	373	382
FU:49	K	CHAR	10		User field 49	383	392
FU:50	K	CHAR	10		User field 50	393	402
FU:51	K	CHAR	20		User field 51	403	422
FU:52	K	CHAR	20		User field 52	423	442
FU:53	K	CHAR	20		User field 53 The shop floor terminal shows this field in the Comment 1 field	443	462
FU:54	K	CHAR	20		User field 54 The shop floor terminal shows this field in the Comment 2 field.	463	482
FU:55	K	CHAR	20		User field 55	483	502
FU:56	K	CHAR	20		User field 56	503	522
FU:57	K	CHAR	20		User field 57	523	542
FU:58	K	CHAR	20		User field 58	543	562
FU:59	K	CHAR	20		User field 59	563	582
FU:60	K	CHAR	20		User field 60	583	602
FU:61	K	CHAR	20		User field 61	603	622
FU:62	K	CHAR	20		User field 62	623	642
FU:63	K	CHAR	20		User field 63	643	662
FU:64	K	CHAR	20		User field 64	663	682
FU:65	K	CHAR	40		User field 65	683	722
FU:66	K	CHAR	40		User field 66	723	762

Specific data for coil-based production

Segment name:	HY72_AG_RF_001_A
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This segment depends on the operation, i.e. the data transferred here populate operation-specific fields in HYDRA. You require this data, if you use the MPL module for coil-based manufacturing.

Field name	V	T	L	D	Description	From	To
ANR	S	CHAR	40		Combined order/OP number	1	40

Field name	V	T	L	D	Description	From	To
RFAGTYP	K	CHAR	1		Special indicator; type of the operation: " " No special processing "P" Packing operation; required in combination with the central weighing function.	41	41
RFABZ	MM	CHAR	1		Identifies mother and child OPs for a planned branch. "M" Mother operation of a planned branch "K" Child OP of a planned branch, which will be confirmed/uploaded in a special goods movement (531) at the goods issue. The indicator should only then be set to "K" if the planned branches must really be confirmed/uploaded as special goods movement.	42	42
RFOPT:RS	MM	CHAR	1		Indicator roll cutting (only relevant if this is a cutting OP) " " No cutting OP "T" Cutting OP with numbering of daughter rolls (batch number has 15 digits: 10 digits+"-"+4-digit index) "M" Cutting OP with numbering of mother rolls; this leads to further mother rolls (batch number has 10 digits).	43	43
RFMANR	MM	CHAR	40		In coil cutting, this reference to the mother OP is used to define the cutting plan. An OP that does <i>not</i> reference to a mother OP must generally refer to itself. Please note: the HYDRA order ID must be configured.	44	83
RFTRANZ	MM	NUMC	5		In case of cutting operations: Number of planned daughter coils per cut. Corresponds to the planned output batches of the operation. If the cutting plan is not defined, 0 is entered here.	84	88
RFTRANZSUM	MM	NUMC	5		For cutting operations (mother OP): number of planned daughter coils per cut (encompassing all branched off material). No specific processing in HYDRA. If the cutting plan is not defined, 0 is entered here.	89	93
RFRANZ	MM	NUMC	6		Planned total number of coils to be produced (mother and daughter coils); no specific processing in HYDRA.	94	99
RFSTKF	MM	NUMC	8		Surface of a piece of the article to be produced. Unit: MM2/ PCE (integer)	100	107
RFBSBRS	MM	DEC	10	3	Total seam width in mm If several coils are produced at the same time in one operation, this field includes the sum total of the separate seam widths. For the planned branches the seam width of the single operation (no summaries) is explicitly set for each operation ("mother" and "child" OPs). Unit: MM	108	119
RFBREITEE	MM	DEC	10	3	Input width of operation Unit: MM	120	131

Field name	V	T	L	D	Description	From	To
RFBREITEA	MM	DEC	10	3	Output width of operation Width of <i>one</i> coil within an operation. For the planned branches the initial width of the single operation (no summaries) is explicitly set for each operation ("mother" and "child" OPs). Unit: MM	132	143
RFAGVFA	MM	DEC	10	3	Mass per unit area Unit: G/M2	144	155
Casing weight	K	DEC	10	3	Specifies the casing weight of the daughter coils for cutting operations. Unit: G	156	167

Order network

Segment name:	HY72_ANETZ_001_A HY72_ANETZ_001_D
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This segment is a global segment used for several orders. Use this segment to define global relationships for several orders. These relationships are respected during planning in the HYDRA shop floor scheduling (HLS) and/or during processing (e.g. target quantity update, if enabled) when the affected operations are planned and/or entered. Data can only be transferred if the transferred orders and/or their operations already exist in HYDRA.

Field	V	T	L	D	Description
Preceding OP	S	CHAR	40		Combined order/ OP number of the predecessor OP
Succeeding OP	S	CHAR	40		Combined order/ OP number of the successor OP
AOB	S	CHAR	2		"ES" by default



Only end/start relations are supported.

You may not transfer relationships between adjacent operations of an order. Relationships are neither supported for split operations and for merged operations and may therefore not be set.



You need the license HLS-KAN if you want to integrate these relationships in the HYDRA Shop Floor Scheduling module (HLS).

Production variants

Segment name:	HY72_FERTVAR_001_A HY72_FERTVAR_001_D
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Use this segment to enter / edit production variants in HYDRA. If configured accordingly , these variants are integrated when identifying production variants as part of the order transfer.

Field	V	T	L	D	Description
Version	S	CHAR	10		Version
Status	M	CHAR	1		Status of the production variant F = Released S = Blocked
Article	S	CHAR	40		Article
Material type	S	CHAR	10		Material type; blank by default (currently not processed)
Machine	S	CHAR	8		Machine
Group	S	CHAR	8		Group
Number (machine)	K	DEC	13	3	Number of machines; "1" by default (currently not processed)
Resource	M	CHAR	40		Resource. Note: The default resource is always assigned the resource type WNR.
Resource family	K	CHAR	18		Resource family (currently not processed)
Number (resource)	K	NUM	8		Number of resources
Target cycle	K	NUM	8		Target cycle in [seconds/ 1000 cycles]
Partitioning	K	DEC	13	3	Partitioning
Setup time	K	NUM	8		Setup time (seconds)
Teardown/retooling time	K	NUM	8		Teardown/retooling time (seconds)
Data ID	SA	CHAR	15		Data ID
Comment	K	CHAR	50		Comment
Priority	M	NUMC	1		Priority
Valid from	K	DATE	10		Valid from (MM/DD/YYYY)
Valid until	K	DATE	10		Valid until (MM/DD/YYYY)

7 Uploading data types: HYDRA --> ERP

Type	Description
CHAR x	Information is left-aligned for the data type CHAR; unnecessary places are filled with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. Numbers are right-aligned; unnecessary places are filled with zeros. Example: "00000002"
DEC_O x.y	Numeric field of the length x and y decimal places. An algebraic sign is preceding the data field ("+" or "-"). Places that are not required are filled with zeros. There is NO DECIMAL SEPARATOR. e.g. DEC_O 13,3: • -1234567890,123 → -1234567890123 • 234567890,3 → +0234567890300
DATE	The date is displayed in the YYYYMMDD format. The field is filled with blanks (if it is not required).
TIME	The time is transferred in the HHMMSS format. The field is populated with "000000".

Generally, HYDRA always transfers a contiguous data structure. Data fields that are not used are filled with blanks. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) and 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, the file transfer may also be performed in the file format that was used until MW 2.0.

8 Operation-related uploads HYDRA --> ERP

Overview

	→→	
	→	

The system uses an individual IDOC with different interface record types (*SATZART* field in the interface record) to transfer the operation-related uploads recorded in HYDRA to the ERP system. There are two different interface record types:

- Uploads based on operations (*SATZART*/record type = "A")

All data records relating to operations are uploaded using the interface record type "A". HYDRA uses a separate entry record type to manage these data records (see e.g. the MOC application [Order-related postings](#)). This indicator allows for the records assigned to the interface record type "A" to be differentiated in more detail. The following entry record types are uploaded:

- Interruption of an operation (postings of the entry record type "U" (*KENN* field in the interface record); is also performed at the end of the shift if the automatic shift function is active).
- Completion of an operation (postings of the entry record type "E" (*KENN* field in the interface record)).

A data record where quantities, processing duration and RPA are not filled out is transferred for each posting relating to orders (interruption, completion) when it comes to operations that are subject to batch management. These data records only transfer the labor utilization of this order. When finishing the order, the data record indicates that the order has actually been finished.

Quantities and times are uploaded in separate data records that are generated in HYDRA while the output batch is changed.

- Personal upload (*SATZART*/record type = "P")

If HYDRA is customized accordingly, you can also upload the "B" records that are managed in HYDRA for the single persons. To do so, use the interface record type "P".

Data is provided at regular intervals (by default approx. every 60 minutes) in the HYDRA subdirectory *./inf_int/interf* (standard system). You can find this directory in the HYDRA directory or in case of multiple system environments in the directory of the system.

Data relevant to operations, such as quantities or times, are posted in the ERP system. The ERP system manages and processes wrong postings separately.

The following specifications result for the IDOC:

Message type:	HY72ADRCK_TT
IDOC type:	HY72ADRCK_TT
File name:	HY72ADRCK_TIMETICKET
File extension	According to configuration in MLE communication (Logical systems > Outbound Configuration File Port) Usually: .dat



It might be the case that negative yield quantities are posted when finishing the operation, provided that part quantities are reported (requires customization services by MPDV) and data is collected at the same time via the total quantity counter on machines working with HYDRA-MDE. Remove the parameter /NEG_MENGE from the myerprck.scr script, provided that the ERP system does not intend to post negative quantities.



An "S" (cancellation) entered in the ERFART field indicates corrected postings. You can find this value in the *Input type* field of the maintenance of postings. Provided that corrections are uploaded, the system uploads the canceled values as absolute values (without algebraic signs), i.e. with an "S" in the ERFART field.

Data structure

Field	T	L	D	Description	From ¹⁵	To ¹⁶
SART	CHAR	1		Interface record type "A" uploads based on operations "P" uploads based on persons	1	1
ERFART	CHAR	1		Origin " " Original data record as recorded "E" data record created manually in the <i>maintenance of postings</i> dialog (edited) "S" Cancellation for ERP	2	2

¹⁵ The indicated number of characters (digits) is calculated based on the exported field length and can be used as reference for HYDRA developers. It does not include the number of characters of the file itself but only of the field SDATA.

¹⁶ See footnote of column "From"

Field	T	L	D	Description	From 15	To ¹⁶
RMNR	CHAR	40		Upload/confirmation number (if available in HYDRA)	3	42
ANR	CHAR	40		Order number The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings.	43	82
AUART	CHAR	5		Order type of the order; according to HYDRA configuration	83	87
STEUER_KZ	CHAR	1		Control indicator of the operation status when being recorded. The indicator depends on the record type of the ADE log record. "L" OP is running (record type "A") "E" OP is finished (record type "E") "U" OP is interrupted (record type "U", "T", "H") Please note: - If you use split OPs, the system uploads/confirms the status of the split master.- The value of this field is not defined for personal uploads.	88	88
AG_STATUS	CHAR	5		Operation status when the ERP upload is being performed, according to the HYDRA configuration.	89	93
SCHICHTNR	NUM	2		Shift according to the shift model assigned to the work center during which the BDE posting was made. The shift number is right-aligned.	94	95
PERSZEIT	DEC_O	10	2	Personal processing time (hours) For postings of the interface record type "A": If this posting record is a U record (interrupt order) or E record (log off order), this field includes the labor time recorded since this record is available. This field includes the logon times of all persons who have logged on to this operation. This field does not include a duration, if it is a T record (reporting part quantities/partial upload), H record (batch posting) or an operation logon. For postings of the interface record type "P": The field includes the length of time a person was logged on to an order. If a person is logged on to several orders/operations simultaneously, the field shows the proportionate labor time per order/operation. The distribution is made proportionally according to the number of operations to which the person is/was logged on at the same time (proportionate labor time).	96	106
				Resource Performance Accounts (RPA) Times are posted to "resource performance accounts". All times refer to the last confirmation/upload performed for the respective operation. The system returns order-related RPAs for postings based on orders (interface record type "A"). The system returns personal RPAs for personal postings (record type "P").		
BMK01	DEC_O	7	2	Resource Performance Account (RPA) 1 in hours	107	114
BMK02	DEC_O	7	2	Resource Performance Account (RPA) 2 in hours	115	122
BMK03	DEC_O	7	2	Resource Performance Account (RPA) 3 in hours	123	130
BMK04	DEC_O	7	2	Resource Performance Account (RPA) 4 in hours	131	138
BMK05	DEC_O	7	2	Resource Performance Account (RPA) 5 in hours	139	146
BMK06	DEC_O	7	2	Resource Performance Account (RPA) 6 in hours	147	154
BMK07	DEC_O	7	2	Resource Performance Account (RPA) 7 in hours	155	162
BMK08	DEC_O	7	2	Resource Performance Account (RPA) 8 in hours	163	170
BMK09	DEC_O	7	2	Resource Performance Account (RPA) 9 in hours	171	178
BMK10	DEC_O	7	2	Resource Performance Account (RPA) 10 in hours	179	186

Field	T	L	D	Description	From 15	To ¹⁶
RPA 11	DEC_O	7	2	Resource Performance Account (RPA) 11 in hours RPA 11 contains the production time (standard use/main utilization time).	187	194
BMK12	DEC_O	7	2	Resource Performance Account (RPA) 12 in hours. RPA 12 contains neutral times (breaks or similar)	195	202
BMK_SUM	DEC_O	7	2	Total of resource performance accounts without breaks (RPA 12) in hours	203	210
FIRMA	CHAR	4		reserved	211	214
GRUPPE	CHAR	8		Group where the workplace/machine is assigned to.	215	222
ARBPL	CHAR	8		Workplace/machine this posting refers to.	223	230
PERSNR	CHAR	10		Personnel number. This data field includes the personnel number of the employee who logged on/off or interrupted the operation. In case of personal postings, this field includes the personnel number of the person for whom data is uploaded/confirmed. The precise length depends on the length configuration of the personnel number in the basic parameter settings of HYDRA.	231	240
LOHNART	CHAR	4		Wage type if stored to the operation. Is directly taken over from the operation .	241	244
GUT_BAS	DEC_O	13	3	Basic quantity of yield, if entered or calculated according to conversion factors.	245	258
AUS_BAS	DEC_O	13	3	Basic quantity of scrap, if entered or calculated according to conversion factors.	259	272
MEINH_BAS	CHAR	3		Basic quantity of the quantity unit, if stored to the operation.	273	275
GUT_PRI	DEC_O	13	3	Collected yield in primary quantity unit Yield recorded in primary quantity unit since the last upload.	276	289
AUS_PRI	DEC_O	13	3	Scrap quantity collected in primary quantity unit since the last upload.	290	303
MEINH_PRI	CHAR	3		Primary unit of entry (primary quantity unit) from the operation.	304	306
				Reasons Reasons are only transferred if the option "confirmation of partial confirmations" is enabled for the order type and only in case of record type "T" postings. Otherwise the fields are empty.		
GUTGR	CHAR	4		Yield reason (deviation reason)	307	310
GUTGR_EXT	CHAR	5		Yield reason (deviation reason) – external reference	311	315
AUSGR	CHAR	4		Scrap reason	316	319
AUSGR_EXT	CHAR	5		Scrap reason – external reference	320	324
ASTATUS	CHAR	1		Order status (control indicator of the order header status) This data field shows the status of an order. When finishing the last recordable operation of an order, this field has the value "E" (end), otherwise "L" (running). The value of this field is not defined for personal confirmations/uploads.	325	325
ANMELD_DAT	DATE	8		Date of the terminal posting (login) in the format YYYYMMDD	326	333
ANMELD_ZEIT	TIME	6		Time of the terminal posting (login) in the format HHMMSS	334	339

Field	T	L	D	Description	From 15	To ¹⁶
ABMELD_DAT	DATE	8		Date of the terminal posting (logoff) in the format YYYYMMDD	340	347
ABMELD_ZEIT	TIME	6		Time of the terminal posting (logoff) in the format HHMMSS	348	353
CHARGE	CHAR	20		Batch number only relevant in connection with ADE-CHV or MPL	354	373
BED_POS	CHAR	10		Entered operator position/function according to the HYDRA configuration, if entered	374	383
LPKZ	CHAR	10		Entered wage/premium indicator according to the HYDRA configuration, if entered	384	393
SOLL_TE	DEC_O	7	2	Target te from the log record (only filled if the LLE-BP license is available)	394	401
SOLL_TR	DEC_O	7	2	Target tr from the log record (only filled if the LLE-BP license is available)	402	409
SOLL_TEB	DEC_O	7	2	Target teb from the log record (only filled if the LLE-BP license is available)	410	417
SOLL_TRB	DEC_O	7	2	Target trb from the log record (only filled if the LLE-BP license is available)	418	425

9 Inbound Transactions

Summary

Menu	System administration → MES Link Enabling (MLE) → Inbound Transactions
Transaction code	intr
Function authorization	intr intr.reset (reset transactions)

Utilization

Inbound transactions provide an overview of the data provided from other systems as well as the result of the inbound processing that follows. The application allows for detailed data to be displayed or new posting processes to be triggered.

Integration

The function allows for data transferred from other systems to be accessed. Other system might be:

- PPS/ERP systems
- Warehouse/material management systems
- Quality management systems

Field Descriptions

Field Descriptions – Inbound Transactions

Transaction number

Unique number that is generated while communicating with the external system.

Status

The status represents the result of the last processing step. The used status are visualized as follows:

Description in the application	Color	Meaning / usage
NEW	Yellow	The record has been provided initially. Further processing still have to take place.
TODO	Yellow	It has been tried already, to

Description in the application	Color	Meaning / usage
		post the record. This attempt has not been successful (for a certain reason) – the record is available to new posting attempt.
REACTIVATED	Orange	A record posted already has been marked for an additional posting attempt.
IN PROCESS	Grey	The record is posted currently.
UNKNOWN	Blue	There is no valid posting routine for the record.
DONE ERROR	Red	The record could not be posted successfully.
DONE	Green	The record could be posted successfully.

IDoc type

IDoc type of the transaction (whether or not the field is filled out, depends on the communication type).

No. of data records

Number of data records included in the transaction

No. of edited DR (data records)

Number of successfully processed data records included in the transaction

No. of unknown data records

Number of unknown data records included in the transaction

No. of erroneous data records

Number of faulty data records included in the transaction (wrong processing)

Time of reception

Date and time when the system received the transaction

Editing time

Date and time when the transaction was edited in the system

IDoc number

IDoc number of transactions (whether or not the field is filled out depends on the communication type).

Message type

Message type of the transaction

Message function

Message function of the transaction

SAP sending port

SAP sending port – only relevant if the communication with SAP is performed via IDoc

SAP sending partner type

SAP sending partner type – only relevant if the communication with SAP is performed via IDocs

SAP sending partner number

SAP sending partner number – only relevant if the communication with SAP is performed via IDocs

Receiver port

Receiver port – only relevant if the communication with SAP is performed via IDocs

Receiver partner type

Receiver partner type – only relevant if the communication with SAP is performed via IDocs

Receiver partner number

Receiver partner number – only relevant if the communication with SAP is performed via IDocs

Reference

Unique database key

Duration

Duration of system processing

Number of attempts

Number of processing attempts

Field Descriptions – Log Table**Application**

Application involved in the processing and editing of the transaction

Log. System

Logical system

Role

Role of the logical system

Designation

Designation of the application

Transaction number

Transaction number that has been processed

Status

Status of processing

Reference

Unique database key

Program

Technical name of the application

Program version

Program version of the application

Program date

Program date of the application

Log file name / log file size

Name and size of the log file

Error file name/error file size

Name and size of the error file

Data file name/data file size

Name and size of the data file

No. of data records

Number of data records included in the transaction

No. of edited data records

Number of successfully processed data records included in the transaction

No. of unknown data records

Number of unknown data records included in the transaction

No. of faulty data records

Number of faulty data records included in the transaction (wrong processing)

Text number

Currently not used

Posting

Currently not used

Created on

Point in time when the entry was created

Toolbar**Reset transaction**

The “reset transaction” button allows for a transaction that has already been processed to be processed again. But this is only possible if the transaction has not yet been archived.

**Data segments**

Data segments may be displayed for a transaction. The system tries to display included application data in relation to individual fields within the data record.

**Log**

Displays the log generated by the respective application.

**Error file**

Displays the error log generated by the respective application.

**Data file**

Displays the data file that is optionally generated by the respective application.

10 Outbound Transactions

Overview

Menu	System Administration → MES Link Enabling (MLE) → Outbound Transactions
Transaction code	outtr
Function authorization	outtr

Purpose

Outbound transactions provide an overview of the data provided to other systems as well as the result of the outbound processing that follows. The application allows for detailed data to be displayed or new posting processes to be triggered.

Integration

The function allows for data transferred to other systems to be accessed. Other system might be:

- PPS/ERP systems
- Warehouse/material management systems
- Quality management systems

Field Descriptions

Field Descriptions – Outbound Transactions

Transaction number

Unique number that is generated while communicating with the external system.

Status

The status represents the result of the last processing step. The used status are visualized as follows:

Description in the application	Color	Meaning / usage
NEW	Yellow	The record has been provided initially. Further processing still have to take place.
TODO	Yellow	It has been tried already, to post the record. This attempt

Description in the application	Color	Meaning / usage
		has not been successful (for a certain reason) – the record is available to new posting attempt.
REACTIVATED	Orange	A record posted already has been marked for an additional posting attempt.
IN PROCESS	Grey	The record is posted currently.
UNKNOWN	Blue	There is no valid posting routine for the record.
DONE ERROR	Red	The record could not be posted successfully.
DONE	Green	The record could be posted successfully.

IDoc type

IDoc type of the transactions (whether or not the field is filled out, depends on the communication type).

No. of data records

Number of data records included in the transaction

No. of edited DR (data records)

Number of successfully processed data records included in the transaction

No. of unknown data records

Number of unknown data records included in the transaction

No. of erroneous data records

Number of faulty data records included in the transaction (wrong processing)

Time of reception

Date and time when the system received the transaction

Editing time

Date and time when the transaction was edited in the system

IDoc number

IDoc number of transactions (whether or not the field is filled out depends on the communication type).

Message type

Message type of the transaction

Message function

Message function of the transaction

SAP sending port

SAP sending port – only relevant if the communication with SAP is performed via IDoc

SAP sending partner type

SAP sending partner type – only relevant if the communication with SAP is performed via IDocs

SAP sending partner number

SAP sending partner number – only relevant if the communication with SAP is performed via IDocs

Receiver port

Receiver port – only relevant if the communication with SAP is performed via IDocs

Receiver partner type

Receiver partner type – only relevant if the communication with SAP is performed via IDocs

Receiver partner number

Receiver partner number – only relevant if the communication with SAP is performed via IDocs

Reference

Unique database key

Duration

Duration of system processing

Number of attempts

Number of processing attempts

Field Descriptions – Log Table**Application**

Application involved in the processing and editing of the transaction

Log. System

Logical system

Role

Role of the logical system

Designation

Designation of the application

Transaction number

Transaction number that has been processed

Status

Status of processing

Reference

Unique database key

Program

Technical name of the application

Program version

Program version of the application

Program date

Program date of the application

Log file name / log file size

Name and size of the log file

Error file name/error file size

Name and size of the error file

Data file name/data file size

Name and size of the data file

No. of data records

Number of data records included in the transaction

No. of edited data records

Number of successfully processed data records included in the transaction

No. of unknown data records

Number of unknown data records included in the transaction

No. of faulty data records

Number of faulty data records included in the transaction (wrong processing)

Text number

Currently not used

Posting

Currently not used

Created on

Point in time when the entry was created

Toolbar**Reset transaction**

Use the “reset transaction” button to process a transaction that has already been processed. This is only possible if the transaction has not yet been archived.

During reactivation, the control record of the transaction is set to the status REACTIVATED. The data records included in the transaction are set to status TODO. Once the data records are transferred and are then included in another transaction, there is no connection between the control record of the original transaction and these data records – you cannot show data records for the original control record any more.

**Data segments**

Data segments may be displayed for a transaction. The system tries to display included application data in relation to individual fields within the data record.

**Log**

Displays the log generated by the respective application.

**Error file**

Displays the error log generated by the respective application.

**Data file**

Displays the data file that is optionally generated by the respective application.

11 Protecting fields of planned operations

Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

Procedure from service pack 12 onwards (b_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	The value includes a condition. Enter the condition, for example, as follows: ANR.ATYP=AG
Active	Yes

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value.
Active	Yes

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

Database field	Values/content
INI_VERWEIS	The value of the VERWEIS column identified from the HYD_INI table via SQL.
SECTION	Section
IDENT	Key
VALUE	Value
BEMERKUNG (comment)	Comment
AKTIV	Active

List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

Acronym	Meaning
ANR.MGRP	Machine group
ANR.MNR	Workplace/ machine
ANR.OPT:PLAN	Planning indicator: M Planned for workplace/machine G Planned for machine group
ANR.DATB	Start date planned (via HLS)
ANR.ZEIB	Start time planned (via HLS)
ANR.DATE	End date planned (via HLS)
ANR.ZEIE	End time planned (via HLS)

Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```

12 Modification to the Order of Uploads

Usage

Subject to the ERP system's way of processing, problems might occur when uploading time tickets to the ERP system. Issue taken: Operations logged on after a shift change are transferred prior to operations interrupted before the change of shifts.

Requirements

The upload program myerprck.exe/out as of version V8.1.1.99 or higher must be used.

Procedure

Add the following entry to the [HYDRA INI configuration](#):

Parameter name	Value
INI name	SAP
Section	TIMETICKET_UPLOAD
Key	ALTERNATIVE_ORDER_BY_CLAUSE
Value	<must remain empty>
Active	Yes
Comment	Changed sorting order

Result

Once activated, uploads will be sorted and reported as follows:

Sort sequence	Comment	Internal number ¹⁷
Logoff date		3
Logoff time		4

¹⁷ Internal number refers to numbering used in the SQL program logic. The number is not relevant to the end user.

Sort sequence	Comment	Internal number ¹⁷
Upload type (record type)	The following order applies for uploading record types: 1. operation logons 2. cancelled operation logoffs 3. cancelled partial uploads, interruptions to operations, batch records (H records) 4. Partial uploads 5. Staff records (B records) 6. interrupted operations, operation logoffs, batch records (H records)	1
Reference		5
Order number		6

13 Test Files

Overview

Attached to this documentation, you will find test files for the interface EIS-ERP. The attachment is only available, if the documentation is in PDF format.

The documentation [Open PDF attachments](#) describes how to call the attached test files.

The following test files are attached to the PDF document:

File	Type	Comment
HY72ADRCK_TIMETICKET.DAT	Outbound processing	Sample file for the operation-related upload HYDRA --> ERP in HYDRA standard format
HY72PPS.DAT	Inbound processing	Sample file for the transfer of order data ERP --> HYDRA in HYDRA standard format